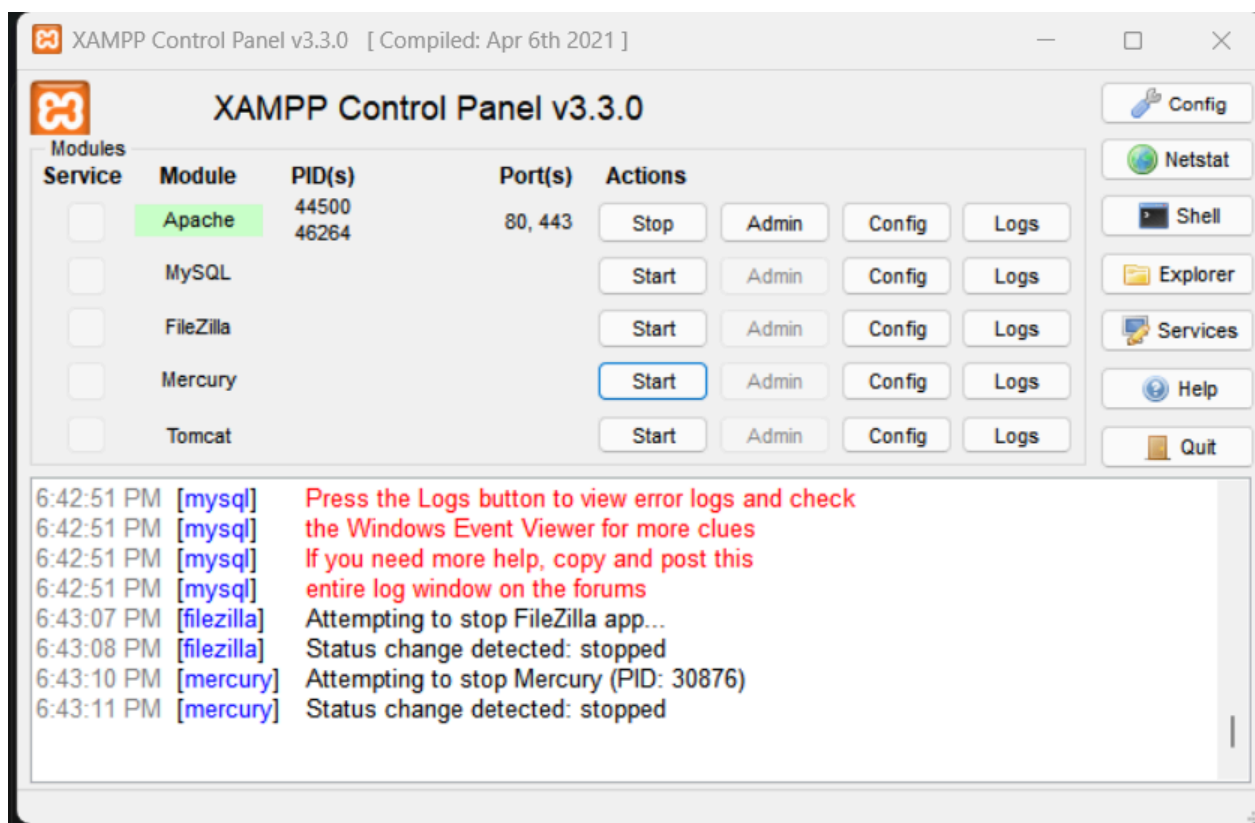


Lab:3 Introduction to Databases and SQL

Exercises

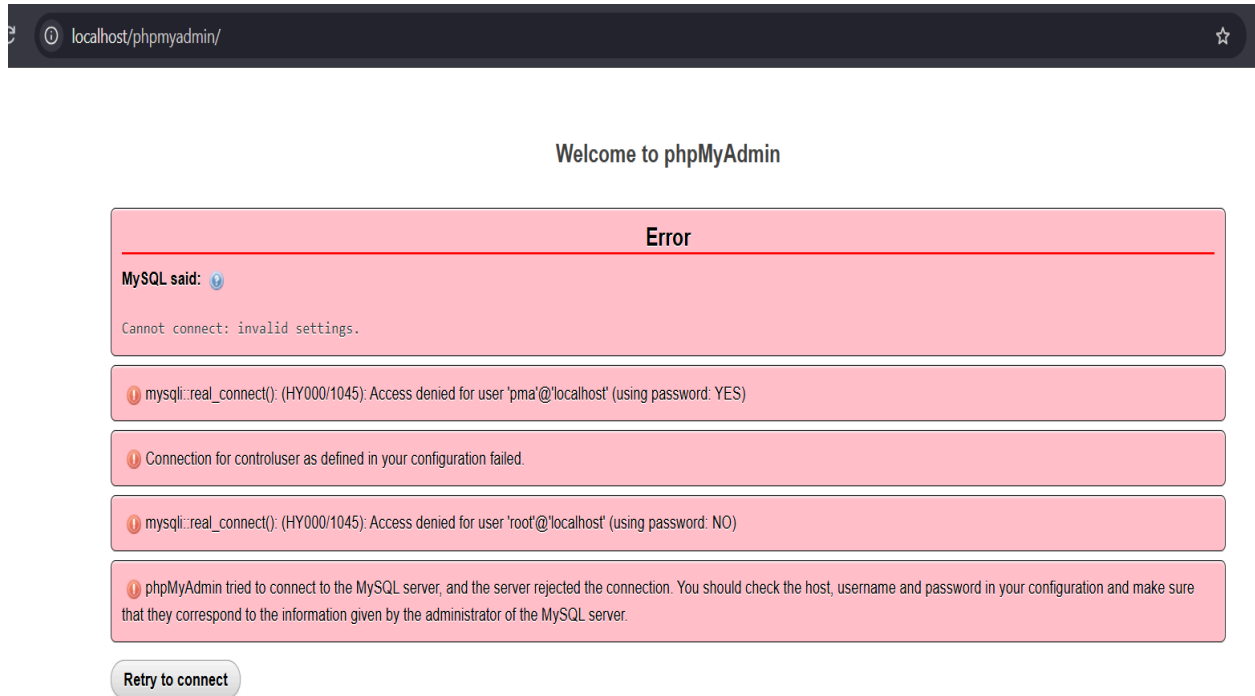
Exercise 1: Setting Up MySQL with XAMPP

1.Started Apache



Apache is successfully running

Image of Error Message



This happened because Port 3306 is already in use from a previous MYSQL already existing on my system. There was no need of starting MYSQL IN XAMPP control panel.

MySQL was configured with root password during the time of installation , but phpMyAdmin is still configured to connect using the old/no password.

How I Fixed This Error:

Step 1

opened the php admin configured file in file explorer:

C:\xampp\phpMyAdmin\config.inc.php

Step 2

Locates:

```
$cfg['Servers'][$i]['password'] = '';  
$cfg['Servers'][$i]['AllowNoPassword'] = true;
```

```
12  
13  /*  
14  | * First server  
15  | */  
16  $i++;  
17  
18  /* Authentication type and info */  
19  $cfg['Servers'][$i]['auth_type'] = 'config';  
20  $cfg['Servers'][$i]['user'] = 'root';  
21  $cfg['Servers'][$i]['password'] = '';  
22  $cfg['Servers'][$i]['extension'] = 'mysqli';  
23  $cfg['Servers'][$i]['AllowNoPassword'] = true;  
24  $cfg['Lang'] = '';  
25  
26  /* Bind to the localhost ipv4 address and tcp */  
27  $cfg['Servers'][$i]['host'] = '127.0.0.1';  
28  $cfg['Servers'][$i]['connect_type'] = 'tcp';
```

nal Ports

Step 3

Changed them to the password used during the time i installed and configured mySQL:

```
$cfg['Servers'][$i]['password'] = 'Princess8686';  
$cfg['Servers'][$i]['AllowNoPassword'] = false;
```

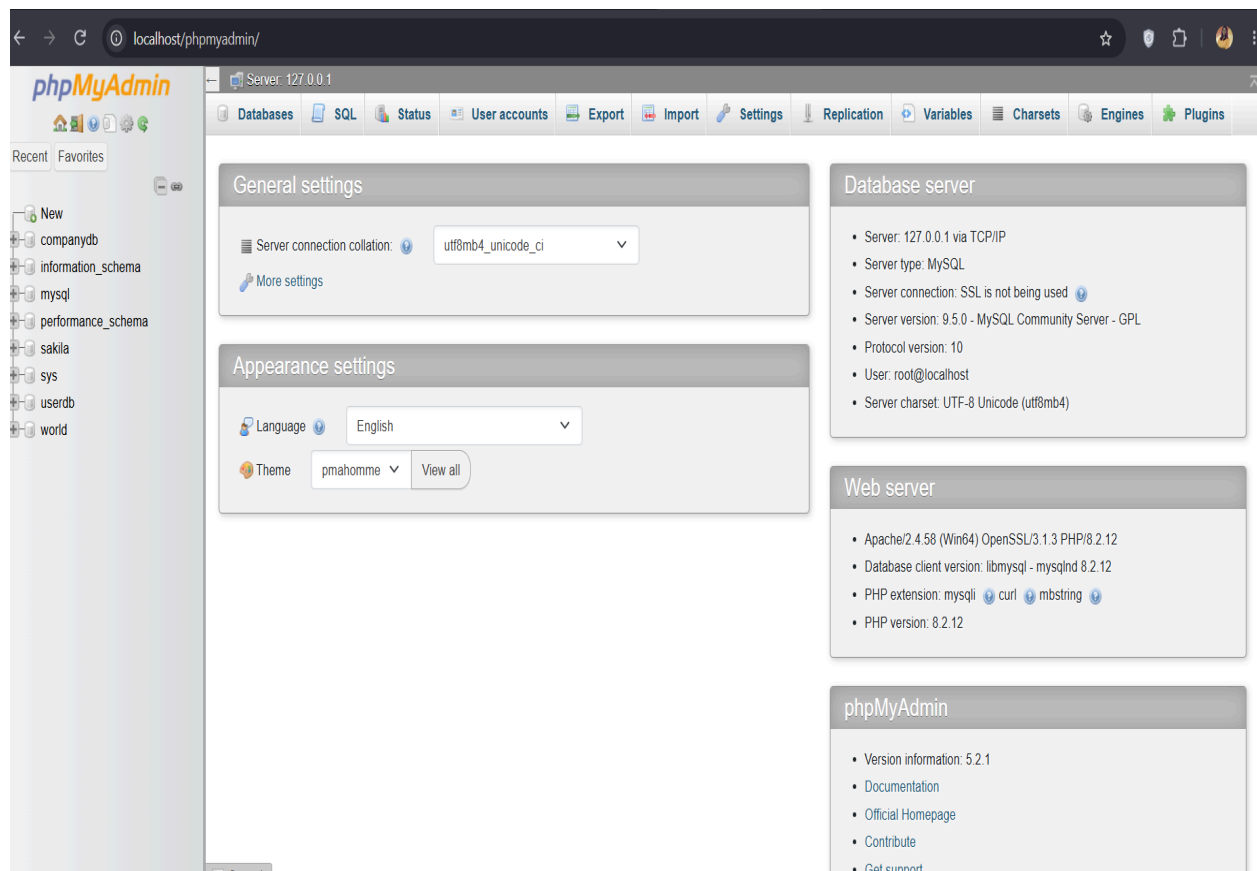
C: > xampp > phpMyAdmin > 🐞 config.inc.php

```
2  /*
4  * cookie
5  */
6  $cfg['blowfish_secret'] = 'xampp'; /* YOU SHOULD CHANGE
7
8  /*
9  * Servers configuration
10 */
11 $i = 0;
12
13 /*
14 * First server
15 */
16 $i++;
17
18 /* Authentication type and info */
19 $cfg['Servers'][$i]['auth_type'] = 'config';
20 $cfg['Servers'][$i]['user'] = 'root';
21 $cfg['Servers'][$i]['password'] = 'Princess8686';|
22 $cfg['Servers'][$i]['extension'] = 'mysqli';
23 $cfg['Servers'][$i]['AllowNoPassword'] = false;
24 $cfg['Lang'] = '';
25
26 /* Bind to the localhost ipv4 address and tcp */
27 $cfg['Servers'][$i]['host'] = '127.0.0.1';
28 $cfg['Servers'][$i]['connect_type'] = 'tcp';
```

2. Access phpMyAdmin

Accessed phpMyAdmin using:

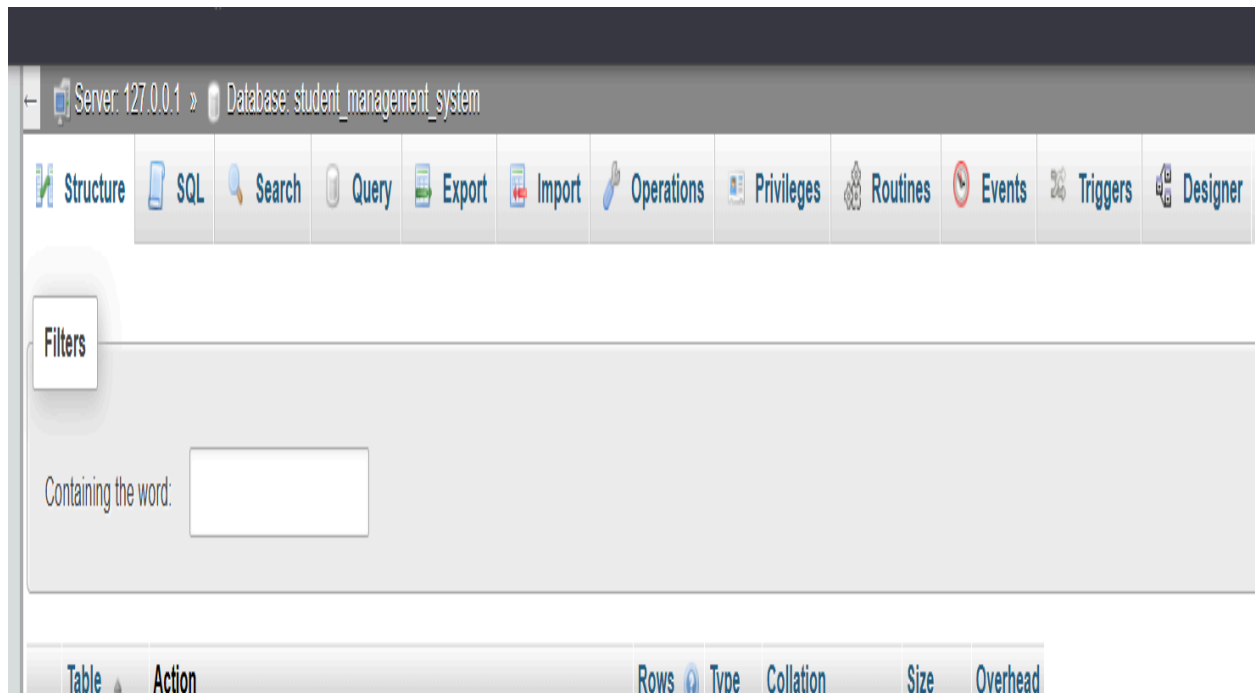
<http://localhost/phpmyadmin/>.



phpMyAdmin successfully opened.

3. Create a New Database

Students_system_management Database



Successfully created students database.

1. **Reflection:** A Student Management System needs a relational database to store and organize information about students, courses, and enrollments. MySQL makes it easy to manage this data, search for information, and connect related records. It also helps keep the data organized and reliable.

Exercise 2: Creating Tables and Defining Columns

Students table

The screenshot shows a database management interface with the following components:

- Navigation Bar:** Server: 127.0.0.1 » Database: student_management_system » Table: students
- Toolbar:** Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, and a dropdown menu.
- Message:** A green banner indicating "MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)"
- SQL Query:** `SELECT * FROM `students``
- Query Options:** A checkbox for "Profiling" and links for "Edit inline", "Edit", "Explain SQL", "Create PHP code", and "Refresh".
- Table Structure:** A table with five columns: `student_id`, `first_name`, `last_name`, `email`, and `date_of_birth`.
- Query Results Operations:** A section containing a "Create view" button.

Course table

Server: 127.0.0.1 » Database: student_management_system » Table: courses

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

Table structure

Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 course_id	int			No	None		AUTO_INCREMENT	Change Drop More
☐	2 course_name	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop More
☐	3 credits	int			No	None			Change Drop More

☐ Check all With selected: Browse Change Drop Primary Unique Index Spatial Fulltext

Print Move columns Normalize

Add column(s)

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	course_id	0	A	No	

Create an index on columns

Enrollment table

localhost/phpmyadmin/index.php?route=/database/structure&server=1&db=student_management_system&table=enrollments

Server: 127.0.0.1 > Database: student_management_system > Table: enrollments

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	enrollment_id	int			No	None		AUTO_INCREMENT	Change Drop More
2	student_id	int			No	None			Change Drop More
3	course_id	int			No	None			Change Drop More
4	enrollment_date	date			No	None			Change Drop More

Check all With selected: Browse Change Drop Primary Unique Index Spatial Fulltext

Print Move columns Normalize

Add 1 column(s) after enrollment_date Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	enrollment_id	0	A	No	

Create an index on 1 columns Go

Successfully created three tables. Students, course and enrollment in the Students_system_management database.

1. **Reflection:** Relationships are created by connecting related columns between tables. For example, the student_id and course_id in the enrollments table connect students to their courses. Foreign keys help maintain data integrity by preventing invalid relationships between records.
-

Exercise 3 & 4: Inserting Data into the Tables and Querying the Database

Students table

Server: 127.0.0.1 » Database: student_management_system » Table: students

[Browse](#) [Structure](#) [SQL](#) [Search](#) [Insert](#) [Export](#) [Import](#) [Privileges](#) [Operations](#)

✓ Showing rows 0 - 3 (4 total, Query took 0.0005 seconds.)

```
SELECT * FROM `students`
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	student_id	first_name	last_name	email	date_of_birth
☐ Edit Copy Delete	1				2019-06-13
☐ Edit Copy Delete	2	Olivil	Chukwuemeka	chukwuemeka@gmail.com	2019-06-13
☐ Edit Copy Delete	3	Jane	Smith	jane.smith@gmail.com	1986-01-20
☐ Edit Copy Delete	4	Emma	Damian	godson1@gmail.com	1992-02-15

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Courses table

Server: 127.0.0.1 » Database: student_management_system » Table: courses

[Browse](#) [Structure](#) [SQL](#) [Search](#) [Insert](#) [Export](#) [Import](#) [Privileges](#) [Operations](#) [Triggers](#)

✓ Showing rows 0 - 2 (3 total, Query took 0.0003 seconds.)

`SELECT * FROM `courses``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	course_id	course_name	credits
☐ Edit Copy Delete	1	Introduction to Databases	3
☐ Edit Copy Delete	2	Cybersecurity Fundamentals	4
☐ Edit Copy Delete	3	Web Development	3

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

Enrollment table

Server: 127.0.0.1 » Database: student_management_system » Table: enrollments

Showing rows 0 - 2 (3 total, Query took 0.0005 seconds.)

`SELECT * FROM `enrollments``

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	enrollment_id	student_id	course_id	enrollment_date
☐ Edit Copy Delete	1	1	2	2024-01-15
☐ Edit Copy Delete	2	2	2	2024-01-17
☐ Edit Copy Delete	3	3	3	2024-01-19

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Successfully inserted values into the students,courses and enrollment Tables and also retrieved records from the Database.

1. Reflection:

SQL allows users to add, view, update, and delete data in a relational database. JOINS make it possible to combine related information from different tables, such as showing a student's name together with the courses they are enrolled in.

Summary

During this lab, I created a relational database called `student_management_system` and created the `students`, `courses`, and `enrollments` tables. I defined primary keys, foreign keys, and relationships between the tables. I also inserted and queried data using SQL. The lab helped me understand how relational databases organize data and how tables can work together.

Potential Improvements

The database can be improved by adding more validation rules, indexes for frequently searched data. Regular backups and proper user access controls can also improve security and performance.